

Two Quadraticizations of a Möbius Tail: LCM Cutoff Return, Generic Row Determinants, and Nonprimitive Endpoint Collapse

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Abstract

Fix $h \neq 0$. TPC-17 reduces a family of symmetric fixed-shift Type-II blocks to the sharp Möbius tail

$$\beta_I(k) = \sum_{\substack{d|k \\ D_0 < d \leq V}} \mu(d), \quad I = (D_0, V],$$

but its separate TT^* theorem concerns an asymmetric packet and cannot be spliced into this tail. We construct two quadraticizations inside the symmetric tail block itself.

The first is an exact least-common-multiple transform

$$\beta_I(k)^2 = \sum_{q|k} \Gamma_I(q), \quad \Gamma_I(q) = \sum_{[d_1, d_2]=q} \mu(d_1)\mu(d_2).$$

Because $V^2 \leq R$, every squared-tail row returns to the original Ramanujan cutoff $q \leq R$. If $D_0 < V/2$, Selberg's classical divisor Gram factorization gives a positive terminal-octave Gram floor and, when $K/V^2 \rightarrow \infty$ as in the explicit TPC-17 profiles, natural k -space L^2 mass; coefficient smallness cannot close it. The second quadraticization first fixes $d \asymp D$, opens $k = dj$, and applies TT^* in j . With $m = \ell d$, $Q = LD$, and $J = K/D$, the determinant is $h(m_1 - m_2)$. We prove by elementary counting that the same-source, small-determinant, and large- (d_1, d_2) channels are all negligible. Thus failure of a tail slice forces a quantitatively large correlation with distinct prime sources, large determinant, and low divisor gcd.

There is a further exact split by $s = (k, h)$. In every nonprimitive sector $s > 1$, writing $H = \text{rad}(s)$ gives

$$\beta_I(k) = \sum_{\substack{e|k \\ (e, H)=1}} \mu(e) \sum_{f|H} \mu(f) \mathbf{1}_{D_0 < ef \leq V},$$

and the inner coefficient is supported only on $(D_0/H, D_0] \cup (V/H, V]$. Hence the whole interior tail cancels pointwise. Moreover, up to a negligible prime-power term, the nonprimitive residual correlation is a finite-model–finite-model correlation governed by three explicit Chinese-remainder conditions.

These results identify a sharper stopping point; they do not estimate the remaining primitive generic correlation. We state a conditional determinant-dispersion input that would close the tail and explain why, under an explicit all-block cover hypothesis for the previous normal form, it reaches the fixed-shift Hardy–Littlewood problem. No twin-prime theorem, new level of distribution, or breach of sieve parity is claimed.

Keywords: Möbius tail; fixed-shift prime correlations; TT^* dispersion; least common multiples; divisor Gram forms; affine determinants; sieve parity.

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1 Introduction and claim ledger

The parity problem is not merely a shortage of numerical cancellation. It is a structural failure of ordinary sieve information to distinguish integers with an odd or even number of prime factors; see Selberg’s classical discussion and modern sieve treatments [4, 12]. The fixed-shift prime-pair problem therefore provides a useful stress test for any proposed spectral or dynamical reformulation: an attractive change of variables is valuable only if every discarded term and every imported distribution theorem is made explicit.

TPC-15–TPC-17 follow this discipline for the finite Ramanujan model [13–15]

$$\Lambda_R(n) = \sum_{q \leq R} \frac{\mu(q)}{\varphi(q)} c_q(n), \quad r_R(n) = \Lambda(n) - \Lambda_R(n).$$

TPC-17 proves cancellation for an assembled initial divisor segment and leaves, on certain symmetric Heath–Brown blocks, the exact coefficient

$$\beta_I(k) = \sum_{\substack{d|k \\ d \in I}} \mu(d), \quad I = (D_0, V].$$

It also proves a TT^* stopping theorem for a different, asymmetric square-root packet. The two statements are not interchangeable: the asymmetric packet has no factor $\beta_I(k)$, while squaring the latter changes both its row geometry and its coefficient class. The first purpose of this paper is to construct the correct quadratic forms directly from the symmetric tail.

1.1 What is proved

There are four unconditional structural statements within the TPC interface.

- (i) The square of the sharp divisor tail has an exact lcm expansion. Since $V^2 \leq R$, its new row label lies back inside the finite-model cutoff. We call this *lcm cutoff return*.
- (ii) If $D_0 < V/2$, the associated divisor Gram form has an explicit positive lower bound coming from $r \in (V/2, V]$. When $K/V^2 \rightarrow \infty$, as in the two explicit TPC-17 profiles, this yields natural k -space L^2 mass. Thus neither Cauchy–Schwarz nor a claim that the tail coefficient is L^2 -small can remove those remaining packets. The Gram identity and terminal-octave mechanism are classical; our contribution is to place them exactly at the TPC-17 stopping gate.
- (iii) A different, d -first TT^* expansion produces row moduli $m = \ell d$ and determinant $h(m_1 - m_2)$. We remove three degenerate families without any distribution theorem. A failed slice must therefore concentrate in the generic channel

$$\ell_1 \neq \ell_2, \quad |m_1 - m_2| > QX^{-\kappa}, \quad (d_1, d_2) \leq X^\kappa.$$

- (iv) Splitting by (k, h) reveals an exact asymmetry. In the primitive sector the two target values have gcd dividing $\ell_1 - \ell_2$. In every nonprimitive sector the interior Möbius interval cancels pointwise and only fixed-ratio endpoint bands remain. Terms containing a target von Mangoldt factor reduce to prime powers, so the surviving correlation is, up to a power-saving error at the relevant scale, a finite-model correlation with an exact CRT kernel.

The individual lcm, Gram, and inclusion–exclusion identities are classical or elementary. The contribution claimed here is their exact coupling to the TPC-17 tail and the resulting stopping decomposition. The combined conclusion is a stopping theorem, not an asymptotic formula. It converts a failed symmetric tail block into either a primitive large-determinant/low-gcd residual witness or a nonprimitive endpoint finite-model witness.

1.2 What is not proved

We do not prove any of the following:

- a fixed- h Hardy–Littlewood asymptotic or infinitely many twin primes;
- cancellation of the primitive generic determinant correlation;
- a new level of distribution for primes;
- that the sharp Möbius interval coefficient is well-factorable;
- that an exponent appearing in an existing one-prime progression theorem can be transplanted to the two-residual correlation here.

Indeed, the last sections explain that a uniform estimate strong enough to remove every relevant generic slice, combined with TPC-16 and TPC-17, would deliver the fixed-shift Hardy–Littlewood conclusion. This is why the remaining statement is recorded as a conditional input rather than advertised as a routine consequence of current dispersion technology.

1.3 Relation to classical and current work

The lcm Gram form belongs to the Selberg-sieve tradition. Closely related Möbius double sums were studied by Dress et al. [3] and de la Bretèche et al. [2]; smoothing and factorization issues are developed systematically by Granville et al. [5]. A recent preprint of Ramaré and Zuniga Altermán [11] makes the same terminal-octave obstruction especially transparent. We therefore make no novelty claim for the bare Gram identity or its endpoint floor.

For distribution beyond the square-root barrier, the relevant lesson from Maynard’s published fixed-residue and well-factorable theorems [9, 10], and from the more recent preprints of Lichtman [7] and Li [6], is that coefficient shape matters as much as the numerical exponent. The sharp interval $\mu(d)\mathbf{1}_I(d)$ contains isolated large primes and is not covered by their balanced factorization hypotheses. Moreover, their conclusions concern one prime in progressions; the object left here is a correlation of two residual affine forms. Kloosterman-fraction estimates such as Bettin and Chandee [1] are important ingredients in dispersion arguments but do not, without a new reduction, estimate our variable-determinant form.

1.4 Organization

Section 2 fixes the inherited block and separates it from TPC-17’s asymmetric packet. Section 3 proves cutoff return. Section 4 proves the coefficient no-go theorem. Section 5 develops the first TT^* witness. Section 6 proves the generic determinant theorem. Section 7 gives the primitive/nonprimitive split and endpoint collapse. Section 8 derives the finite-model CRT kernel. Section 9 assembles the stopping theorem and states the exact conditional closure gate. Section 10 describes the finite certificate and summarizes the boundary.

2 The symmetric-tail interface

Fix $h \in \mathbb{Z} \setminus \{0\}$ and $0 < \delta < 1/2$. Throughout,

$$R = \lfloor X^{1/2-\delta} \rfloor, \quad V = \lfloor R^{1/2} \rfloor. \quad (1)$$

Let $W \in C_c^\infty((0, \infty); \mathbb{R})$ and let $\psi \in C_c^\infty([1/2, 2]; [0, 1])$. We use the finite model

$$\Lambda_R(n) = \sum_{q \leq R} \frac{\mu(q)}{\varphi(q)} c_q(n), \quad r_R(n) = \Lambda(n) - \Lambda_R(n), \quad (2)$$

where c_q is the Ramanujan sum. The elementary divisor expansion used in TPC-16 [14] gives, uniformly for $n \asymp X$ and every $\varepsilon > 0$,

$$|\Lambda_R(n)| + |r_R(n)| \ll_{\varepsilon, h, W, \delta} X^\varepsilon. \quad (3)$$

Let $D_0 < V$ and put

$$I = (D_0, V] \cap \mathbb{N}, \quad \beta_I(k) = \sum_{\substack{d|k \\ d \in I}} \mu(d). \quad (4)$$

For dyadic L, K satisfying

$$LK \asymp_W X, \quad L > 2R, \quad K > 2V, \quad (5)$$

define the symmetric tail block

$$\mathcal{T}_{L,K}(I) = \sum_{\ell, k \geq 1} \Lambda(\ell) \beta_I(k) r_R(\ell k + h) W(\ell k / X) \psi(\ell / L) \psi(k / K). \quad (6)$$

The lower conditions $\ell > R$ and $k > V$ are automatic on the support. This is precisely the tail normal form furnished by the assembled-prefix theorem of TPC-17 on each block satisfying its published or explicitly version-locked distribution hypotheses [15].

Remark 2.1 (No asymmetric splice). TPC-17 also studies an asymmetric packet with coefficient 1 in the k variable. Its TT^* theorem does not apply to (6): replacing 1 by $\beta_I(k)$ changes the squared weight and creates an lcm row label. Every quadratic form below is derived anew from (6).

Let $\mathcal{T}_{L,K}^{\mathbb{P}}(I)$ be the restriction of (6) to prime ℓ , with source weight $\log \ell$. The source prime powers are negligible before either quadraticization.

Lemma 2.2 (Prime-source reduction). *For every fixed $\varepsilon > 0$,*

$$\mathcal{T}_{L,K}(I) - \mathcal{T}_{L,K}^{\mathbb{P}}(I) \ll_{\varepsilon, h, W, \psi, \delta} XL^{-1/2} X^\varepsilon. \quad (7)$$

In particular this has arbitrary logarithmic saving when $L \geq X^{\lambda_0}$ for some fixed $\lambda_0 > 0$.

Proof. The elementary bound $|\beta_I(k)| \leq \tau(k) \ll_\varepsilon X^\varepsilon$ holds on the support. The total von Mangoldt weight of prime powers $p^a \asymp L$, $a \geq 2$, is $O_\varepsilon(L^{1/2+\varepsilon})$. For each such source there are $O_W(K)$ possible k , and (3) applies. Hence the omitted contribution is

$$\ll L^{1/2+\varepsilon} K X^{2\varepsilon} \ll_W XL^{-1/2} X^{3\varepsilon},$$

after using $LK \asymp_W X$ and renaming 3ε . □

We henceforth write

$$\lambda_\ell = (\log \ell) \mathbf{1}_{\ell \text{ prime}} \psi(\ell / L), \quad A_{\ell,k} = r_R(\ell k + h) W(\ell k / X) \psi(k / K), \quad (8)$$

so that

$$\mathcal{T}_{L,K}^{\mathbb{P}}(I) = \sum_k \beta_I(k) S_k, \quad S_k = \sum_\ell \lambda_\ell A_{\ell,k}. \quad (9)$$

All weights and subsequent correlations are real. Absolute values are retained in inequalities only for estimates.

3 The lcm square map and cutoff return

Define the lcm coefficient

$$\Gamma_I(q) = \sum_{\substack{d_1, d_2 \in I \\ [d_1, d_2] = q}} \mu(d_1)\mu(d_2). \quad (10)$$

Theorem 3.1 (Exact lcm transform). *For every $k \geq 1$,*

$$\boxed{\beta_I(k)^2 = \sum_{q|k} \Gamma_I(q).} \quad (11)$$

Consequently,

$$\Gamma_I(q) = \sum_{a|q} \mu(q/a) \beta_I(a)^2. \quad (12)$$

Moreover,

$$\text{supp } \Gamma_I \subset \{q \leq V^2 : q \text{ squarefree}\}, \quad |\Gamma_I(q)| \leq 3^{\omega(q)}, \quad (13)$$

and

$$\sum_q |\Gamma_I(q)| \leq V^2. \quad (14)$$

Proof. Expand the square in (4). The conditions $d_1 \mid k$ and $d_2 \mid k$ are equivalent to $[d_1, d_2] \mid k$; grouping by that least common multiple proves (11). Equation (12) is ordinary Möbius inversion.

Only squarefree d_i contribute. Hence their lcm is squarefree and at most V^2 . For a squarefree q , each prime divisor can lie in d_1 only, d_2 only, or both, giving at most $3^{\omega(q)}$ ordered pairs. Finally, summing absolute values before grouping by the lcm gives at most the number of ordered pairs $d_1, d_2 \in I$, which is at most V^2 . $\square$

The sign structure becomes more transparent after separating the gcd.

Proposition 3.2 (Pairwise-coprime lcm coordinates). *One has the exact formula*

$$\Gamma_I(q) = \sum_{\substack{gab=q \\ g, a, b \text{ squarefree} \\ (g, a) = (g, b) = (a, b) = 1 \\ D_0 < ga \leq V, D_0 < gb \leq V}} \mu(a)\mu(b), \quad (15)$$

The squarefree condition is part of the summation range.

Proof. For a nonzero pair in (10), put $g = (d_1, d_2)$, $a = d_1/g$, and $b = d_2/g$. Squarefreeness makes g, a, b pairwise coprime, $[d_1, d_2] = gab$, and

$$\mu(d_1)\mu(d_2) = \mu(ga)\mu(gb) = \mu(g)^2\mu(a)\mu(b) = \mu(a)\mu(b).$$

The construction is reversible. $\square$

Since $V^2 \leq R$ by (1), the lcm label created by squaring never leaves the finite Ramanujan cutoff:

$$\boxed{q = [d_1, d_2] \leq V^2 \leq R.} \quad (16)$$

This is an algebraic support statement, not a cancellation estimate. In particular, the diagonal pair $d_1 = d_2 = q$ contributes

$$\Gamma_I(q) = \mu(q)^2 \mathbf{1}_{q \in I} + \Gamma_I^\neq(q), \quad (17)$$

and neither term on the right is automatically small.

For later use, summing (11) against a test function F gives the exact row assembly

$$\sum_k \beta_I(k)^2 F(k) = \sum_{q \leq R} \Gamma_I(q) \sum_j F(qj). \quad (18)$$

Thus the same- k quadraticization has a small row label $q \leq R$. The d -first quadraticization of section 6 instead has the large row label $\ell d > R$. These are complementary normal forms, not two notations for the same modulus.

4 Divisor Gram positivity and the terminal-octave floor

For a finitely supported sequence $u = (u_d)$, define

$$\mathcal{Q}(u) = \sum_{d,e} \frac{u_d \overline{u_e}}{[d,e]}. \quad (19)$$

Lemma 4.1 (Selberg divisor Gram factorization). *For every finitely supported u ,*

$$\boxed{\mathcal{Q}(u) = \sum_{r \geq 1} \varphi(r) \left| \sum_{r|d} \frac{u_d}{d} \right|^2 \geq 0.} \quad (20)$$

Proof. Use

$$\frac{1}{[d,e]} = \frac{(d,e)}{de} = \frac{1}{de} \sum_{r|(d,e)} \varphi(r)$$

and interchange the finite sums. □

Take $u_d = \mu(d) \mathbf{1}_{d \in I}$ and write

$$Q_I = \sum_{d,e \in I} \frac{\mu(d)\mu(e)}{[d,e]}. \quad (21)$$

Theorem 4.2 (Terminal-octave rigidity). *If $D_0 < V/2$, then*

$$Q_I \geq \sum_{V/2 < r \leq V} \frac{\mu(r)^2 \varphi(r)}{r^2} \quad (22)$$

$$= C_{\text{sf}} \log 2 + o(1), \quad C_{\text{sf}} = \prod_p \left(1 - \frac{2}{p^2} + \frac{1}{p^3} \right) > 0. \quad (23)$$

Proof. In (20), retain only $V/2 < r \leq V$. The only multiple of such an r not exceeding V is $d = r$; because $D_0 < V/2$, it belongs to I . This proves (22).

For completeness, put $a(n) = \mu(n)^2 \varphi(n)/n$. Its mean value is

$$\prod_p (1 - p^{-1}) \left(1 + \frac{1 - p^{-1}}{p} \right) = \prod_p (1 - 2p^{-2} + p^{-3}) = C_{\text{sf}}.$$

Standard mean-value theory for multiplicative functions, followed by partial summation, gives

$$\sum_{y < r \leq z} \frac{a(r)}{r} = C_{\text{sf}} \log(z/y) + o(1)$$

when z/y is fixed and $y \rightarrow \infty$. Take $y = V/2$, $z = V$. □

The finite- K meaning of this Gram mass is exact up to an elementary lattice error.

Proposition 4.3 (Weighted second moment). *Let $w \in C_c^1((0, \infty))$. Then*

$$\sum_{k \geq 1} \beta_I(k)^2 w(k/K) = K I_0(w) Q_I + O_w(V^2), \quad I_0(w) = \int_0^\infty w(t) dt. \quad (24)$$

In particular, if $w \geq 0$, $I_0(w) > 0$, $D_0 < V/2$, and $K/V^2 \rightarrow \infty$, then

$$\sum_k \beta_I(k)^2 w(k/K) \gg_w K. \quad (25)$$

Proof. Insert (11) and use

$$\sum_j w(qj/K) = \frac{K}{q} I_0(w) + O_w(1).$$

The main coefficient is

$$\sum_q \frac{\Gamma_I(q)}{q} = \sum_{d,e \in I} \frac{\mu(d)\mu(e)}{[d,e]} = Q_I,$$

while (14) bounds the total error by $O_w(V^2)$. The final assertion follows from theorem 4.2. $\square$

Remark 4.4 (A rigorous coefficient-shape obstruction). Suppose $D_0 < \sqrt{V}$ and let $p \in (\sqrt{V}, V]$ be prime. Then $\mu(p)\mathbf{1}_I(p) = -1$. On the other hand, if a and b are both supported on $[1, \sqrt{V}]$, then $(a * b)(p) = 0$. Hence the exact sharp coefficient $\mu(d)\mathbf{1}_I(d)$ cannot be a finite linear combination of such balanced convolutions. This elementary observation does not rule out every notion of factorability, but it does rule out the balanced shape needed for a direct invocation of the current well-factorable distribution theorems. Together with (25), it shows why neither a formal factor split nor coefficient L^2 smallness closes the tail.

The identities in this section are classical Selberg diagonalization. The endpoint floor has close precedents in de la Bretèche et al. [2], Dress et al. [3], Ramaré and Zuniga Altermán [11]. The new role here is diagnostic: it identifies which tempting route at the TPC-17 tail gate is unavailable before any prime-correlation estimate is attempted.

5 The same- k quadraticization

We first apply Cauchy–Schwarz without opening a divisor. With S_k as in (9), define

$$\mathcal{D}_I := \sum_{\ell, k} \beta_I(k)^2 \lambda_\ell^2 |A_{\ell, k}|^2, \quad (26)$$

$$\mathcal{C}_I^{\text{off}} := \sum_{\ell_1 \neq \ell_2} \lambda_{\ell_1} \lambda_{\ell_2} \sum_k \beta_I(k)^2 A_{\ell_1, k} A_{\ell_2, k}. \quad (27)$$

Theorem 5.1 (Tail-weighted TT^* stopping theorem). *Assume (5) and $L \geq X^{\lambda_0}$ for a fixed $\lambda_0 > 0$. Then*

$$|\mathcal{T}_{L, K}^{\mathbb{P}}(I)|^2 \ll_W K(\mathcal{D}_I + \mathcal{C}_I^{\text{off}}), \quad \mathcal{D}_I \ll_\varepsilon X^{1+\varepsilon}. \quad (28)$$

Consequently, for fixed $a \geq 0$ and $\eta > 0$,

$$|\mathcal{T}_{L, K}^{\mathbb{P}}(I)| \geq \frac{\eta X}{(\log X)^a} \implies \mathcal{C}_I^{\text{off}} \gg_{\eta, W} \frac{XL}{(\log X)^{2a}} \quad (29)$$

for all sufficiently large X .

Proof. There are $O_W(K)$ integers on the k support, so

$$\left| \sum_k \beta_I(k) S_k \right|^2 \ll_W K \sum_k \beta_I(k)^2 |S_k|^2.$$

Expanding the last square gives exactly $\mathcal{D}_I + \mathcal{C}_I^{\text{off}}$. The bounds $|\beta_I(k)| \ll_\varepsilon X^\varepsilon$, $|A_{\ell,k}| \ll_\varepsilon X^\varepsilon$, and $\lambda_\ell \ll \log X$, together with $LK \asymp_W X$, give $\mathcal{D}_I \ll_\varepsilon X^{1+\varepsilon}$ after renaming ε .

If the hypothesis in (29) holds, Cauchy's lower bound and $K \asymp_W X/L$ give

$$\mathcal{D}_I + \mathcal{C}_I^{\text{off}} \gg_{\eta,W} \frac{XL}{(\log X)^{2a}}.$$

Choose $\varepsilon < \lambda_0/2$. Then the diagonal is smaller than the right-hand side by a fixed power of X , proving the claim. $\square$

Insert the lcm transform and write $k = qj$. Equation (27) becomes

$$\begin{aligned} \mathcal{C}_I^{\text{off}} = & \sum_{\ell_1 \neq \ell_2} \lambda_{\ell_1} \lambda_{\ell_2} \sum_{q \leq R} \Gamma_I(q) \sum_{j \geq 1} r_R(\ell_1 q j + h) r_R(\ell_2 q j + h) \\ & \times W(\ell_1 q j / X) W(\ell_2 q j / X) \psi(q j / K)^2. \end{aligned} \quad (30)$$

For fixed q, ℓ_1, ℓ_2 , the two affine forms in j have displayed determinant

$$\det \begin{pmatrix} \ell_1 q & h \\ \ell_2 q & h \end{pmatrix} = h q (\ell_1 - \ell_2). \quad (31)$$

In the original variable k , however, the transverse determinant is $h(\ell_1 - \ell_2)$; q records the separate row condition $q \mid k$. This distinction reappears exactly in the CRT conditions of section 8.

The g, a, b coordinates of proposition 3.2 give the fully resolved version

$$\begin{aligned} \mathcal{C}_I^{\text{off}} = & \sum_{\substack{g,a,b \\ g,a,b \text{ squarefree} \\ (g,a)=(g,b)=(a,b)=1 \\ D_0 < ga, gb \leq V}} \mu(a) \mu(b) \sum_{\ell_1 \neq \ell_2} \lambda_{\ell_1} \lambda_{\ell_2} \\ & \times \sum_j r_R(\ell_1 g a b j + h) r_R(\ell_2 g a b j + h) \mathcal{W}_{\ell_1, \ell_2}(g a b j), \end{aligned} \quad (32)$$

where

$$\mathcal{W}_{\ell_1, \ell_2}(k) = W(\ell_1 k / X) W(\ell_2 k / X) \psi(k / K)^2.$$

Corollary 5.2 (Dyadic lcm witnesses). *Under the failure hypothesis in (29), one of the $O(\log^2 X)$ signed cells*

$$q \asymp Q_0, \quad |\ell_1 - \ell_2| \asymp \Delta$$

in (30) has positive contribution

$$\gg_{\eta,W} \frac{XL}{(\log X)^{2a+2}}. \quad (33)$$

At the completely factor-resolved level, one of the $O(\log^4 X)$ cells

$$g \asymp G, \quad a \asymp A, \quad b \asymp B, \quad |\ell_1 - \ell_2| \asymp \Delta$$

has positive contribution

$$\gg_{\eta,W} \frac{XL}{(\log X)^{2a+4}}. \quad (34)$$

Proof. Use [theorem 5.1](#) and partition the positive total (27) into dyadic signed subsums. The triangle inequality bounds the number of cells by the stated powers of $\log X$; hence at least one cell is positive and carries at least the average share. Empty endpoint ranges are omitted. $\square$

The lcm route therefore returns every squared coefficient to a small row $q \leq R$ and produces a detailed witness. It does not by itself separate the elementary determinant degeneracies. For that purpose we now open one divisor before applying TT^* .

6 The opened- d dispersion form

Let ω_D be a bounded smooth dyadic cutoff supported on $[D, 2D]$, with $1 \leq D \leq V$. When slices are reassembled, the ω_D are taken from one fixed bounded-overlap partition of unity on I , satisfying

$$\sum_D \omega_D(d) = 1 \quad (d \in I), \quad \sup_D D^r \|\omega_D^{(r)}\|_\infty < \infty \quad (r \geq 0). \quad (35)$$

A d -resolved tail slice is

$$\mathcal{T}_D = \sum_{j \geq 1} \sum_{d \in I} \mu(d) \omega_D(d) \sum_\ell \lambda_\ell r_R(\ell d j + h) W(\ell d j / X) \psi(d j / K) \xi(d, j), \quad (36)$$

where $|\xi(d, j)| \leq 1$. The admissible masks are fixed real-valued functions of (d, j) modulo $|h|$, from a family of complexity $O_h(1)$ and independent of X , r_R , and the source variables. This includes the constant mask $\xi \equiv 1$ and the primitive mask $(d j, h) = 1$. Put

$$J = K/D, \quad Q = LD. \quad (37)$$

On the support, $j \asymp J$, $m = \ell d \asymp Q$, and

$$QJ \asymp_W X. \quad (38)$$

For a row label $\alpha = (\ell, d)$, write $m_\alpha = \ell d$ and absorb all bounded cutoffs into $B_\alpha(j)$. Expanding after Cauchy in j gives

$$|\mathcal{T}_D|^2 \ll_W J(\mathcal{E}_D + \mathcal{C}_D^{\text{off}}), \quad (39)$$

$$\mathcal{E}_D := \sum_j \sum_\alpha \mu(d)^2 (\log \ell)^2 r_R(m_\alpha j + h)^2 |B_\alpha(j)|^2, \quad (40)$$

$$\mathcal{C}_D^{\text{off}} := \sum_j \sum_{\alpha_1 \neq \alpha_2} \mu(d_1) \mu(d_2) (\log \ell_1) (\log \ell_2) r_R(m_1 j + h) r_R(m_2 j + h) B_{\alpha_1}(j) B_{\alpha_2}(j). \quad (41)$$

Here and below the prime and support restrictions from (36) are understood.

Because $\ell \geq L/2 > R \geq V \geq d_1, d_2$, unique factorization gives

$$\ell_1 d_1 = \ell_2 d_2 \iff (\ell_1, d_1) = (\ell_2, d_2). \quad (42)$$

Thus every off-diagonal pair has $m_1 \neq m_2$, and its two affine forms have determinant

$$\boxed{\det \begin{pmatrix} m_1 & h \\ m_2 & h \end{pmatrix} = h(m_1 - m_2) = h(\ell_1 d_1 - \ell_2 d_2).} \quad (43)$$

For distinct sources the same size separation gives the exact identity

$$(m_1, m_2) = (d_1, d_2). \quad (44)$$

Lemma 6.1 (Diagonal energy). *For every $\varepsilon > 0$,*

$$\mathcal{E}_D \ll_{\varepsilon, h, W, \psi} X^{1+\varepsilon}. \quad (45)$$

Proof. There are $O(LDJ) = O(X)$ triples (ℓ, d, j) on the support. Apply (3), the boundedness of the cutoffs, and absorb the logarithmic source weight into X^ε . $\square$

We next partition the off-diagonal pairs. Fix $\kappa > 0$ and let $\mathcal{C}_D^{\text{same}}$, $\mathcal{C}_D^{\text{near}}$, $\mathcal{C}_D^{\text{gcd}}$, and $\mathcal{C}_D^{\text{gen}}$ be the signed subsum of (41) on, respectively,

$$\mathcal{S} = \{\ell_1 = \ell_2, d_1 \neq d_2\}, \quad (46)$$

$$\mathcal{N} = \{\ell_1 \neq \ell_2, 0 < |m_1 - m_2| \leq QX^{-\kappa}\}, \quad (47)$$

$$\mathcal{G} = \{\ell_1 \neq \ell_2, |m_1 - m_2| > QX^{-\kappa}, (d_1, d_2) > X^\kappa\}, \quad (48)$$

$$\mathcal{Gen} = \{\ell_1 \neq \ell_2, |m_1 - m_2| > QX^{-\kappa}, (d_1, d_2) \leq X^\kappa\}. \quad (49)$$

These four sets form a disjoint partition.

Theorem 6.2 (Elementary removal of the degenerate channels). *For every fixed $\varepsilon > 0$,*

$$|\mathcal{C}_D^{\text{same}}| \ll_\varepsilon XQ L^{-1} X^\varepsilon, \quad (50)$$

$$|\mathcal{C}_D^{\text{near}}| \ll_\varepsilon XQ(X^{-\kappa} + L^{-1})X^\varepsilon, \quad (51)$$

$$|\mathcal{C}_D^{\text{gcd}}| \ll_\varepsilon XQ X^{-\kappa+\varepsilon}. \quad (52)$$

If $L \geq X^{\lambda_0}$ for fixed $\lambda_0 > 0$, all three have a fixed power saving after choosing ε sufficiently small. In particular they are $O_A(XQ(\log X)^{-A})$ for every fixed $A > 0$.

Proof. By (3), each summand is $O_\varepsilon(X^\varepsilon)$ after absorbing logarithms. It therefore suffices to count tuples.

For $\mathcal{S}$, there are $O(LD^2J)$ choices, which equals $O(XQ/L)$ because $Q = LD$ and $QJ \asymp X$.

For $\mathcal{N}$, fix ℓ_1, d_1, d_2 . The condition on $\ell_2 d_2$ confines ℓ_2 to an interval containing $O(QX^{-\kappa}/D + 1) = O(LX^{-\kappa} + 1)$ integers. Thus the number of row pairs is

$$O(LD^2(LX^{-\kappa} + 1)) = O(Q^2X^{-\kappa} + LD^2).$$

Multiplication by $O(J)$ possible j gives $O(XQ(X^{-\kappa} + L^{-1}))$.

Finally,

$$\#\{d_1, d_2 \asymp D : (d_1, d_2) > X^\kappa\} \ll \sum_{g > X^\kappa} (D/g)^2 \ll D^2 X^{-\kappa}.$$

There are $O(L^2)$ source pairs and $O(J)$ values of j , giving (52). The final assertion follows by taking the soft pointwise exponent smaller than a fixed fraction of $\min(\kappa, \lambda_0)$. $\square$

Theorem 6.3 (Generic determinant stopping theorem). *Fix $a \geq 0$, $\eta > 0$, $\kappa > 0$, and $\lambda_0 > 0$. Under (5), suppose $L \geq X^{\lambda_0}$ and*

$$|\mathcal{T}_D| \geq \frac{\eta X}{(\log X)^a}. \quad (53)$$

Then, for sufficiently large X ,

$$\boxed{\mathcal{C}_D^{\text{gen}} \gg_{\eta, W} \frac{XQ}{(\log X)^{2a}}.} \quad (54)$$

Thus a failed slice forces distinct prime sources, determinant $|h(m_1 - m_2)| \gg_h QX^{-\kappa}$, and divisor gcd at most X^κ .

Proof. By (39), (53), and $J \asymp X/Q$,

$$\mathcal{E}_D + \mathcal{C}_D^{\text{off}} \gg_{\eta, W} \frac{XQ}{(\log X)^{2a}}.$$

The diagonal bound is smaller by a fixed power because $Q \geq L$. The three degenerate pieces are smaller by [theorem 6.2](#). Since their signed sum together with $\mathcal{C}_D^{\text{gen}}$ equals $\mathcal{C}_D^{\text{off}}$, the displayed lower bound remains in the generic channel. $\square$

This is an unconditional narrowing of the obstruction. It is not an estimate for $\mathcal{C}_D^{\text{gen}}$; the residual signs in that channel remain the hard arithmetic input.

7 Primitive content and nonprimitive endpoint collapse

The determinant is only part of the common-divisor geometry. For a given k , put

$$s = (k, h), \quad k = sk', \quad h = sh', \quad (k', h') = 1. \quad (55)$$

Proposition 7.1 (Target gcd geometry). *For distinct source primes ℓ_1, ℓ_2 there exists $t \mid (\ell_1 - \ell_2)$ such that*

$$\gcd(\ell_1 k + h, \ell_2 k + h) = st. \quad (56)$$

In particular, on the primitive sector $(k, h) = 1$,

$$\gcd(\ell_1 k + h, \ell_2 k + h) \mid |\ell_1 - \ell_2|. \quad (57)$$

Proof. After removing s , the two target values are $\ell_i k' + h'$. Their gcd divides their difference $(\ell_1 - \ell_2)k'$. It is coprime to k' , because $\gcd(\ell_i k' + h', k') = (h', k') = 1$. Hence it divides $\ell_1 - \ell_2$. Restoring s proves the formula. $\square$

For the lcm row $k = qj$, knowing (q, h) alone does not determine the content. Indeed, with $s_0 = (q, h)$,

$$(qj, h) = s_0 \gcd\left(j, \frac{h}{s_0}\right). \quad (58)$$

Thus a complete content split must also separate the finitely many divisors of h/s_0 . Since h is fixed, this costs no power of $\log X$.

The Möbius coefficient undergoes an exact cancellation on every nonprimitive sector.

Theorem 7.2 (Nonprimitive endpoint collapse). *Let $s = (k, h) > 1$ and $H = \text{rad}(s)$. Then*

$$\boxed{\beta_I(k) = \sum_{\substack{e \mid k \\ (e, H) = 1}} \mu(e) \Omega_H(e), \quad \Omega_H(e) = \sum_{f \mid H} \mu(f) \mathbf{1}_{D_0 < ef \leq V}.} \quad (59)$$

Moreover,

$$\boxed{\text{supp } \Omega_H \subset (D_0/H, D_0] \cup (V/H, V].} \quad (60)$$

Thus the full interior $D_0 < e \leq V/H$ cancels pointwise.

Proof. Since $H \mid k$, every squarefree divisor $d \mid k$ has a unique factorization $d = ef$ with $f \mid H$ and $(e, H) = 1$. The two factors are coprime, so $\mu(d) = \mu(e)\mu(f)$. Inserting the interval condition on d proves (59).

If $e \leq D_0/H$, then $ef \leq D_0$ for every $f \mid H$, so $\Omega_H(e) = 0$. If $D_0 < e \leq V/H$, every $f \mid H$ satisfies $D_0 < ef \leq V$, and

$$\Omega_H(e) = \sum_{f \mid H} \mu(f) = 0$$

because $H > 1$. Finally $e > V$ gives no admissible f . The only remaining possibilities lie in the two intervals in (60). The intervals may overlap, which causes no problem. $\square$

Because $s \mid h$, one has $H \leq |h|$. Hence each surviving interval has a fixed multiplicative width depending only on h . This is a strict pointwise identity, not an average cancellation statement. After squaring (59), every nonprimitive $\beta_I(k)^2$ is assembled from two endpoint divisor labels e_1, e_2 in the same fixed union.

We now show that a nonprimitive residual pair contains no genuinely new target-prime correlation. For $s \mid h$, define

$$\mathcal{C}_{I,s}^{\text{off}} := \sum_{\ell_1 \neq \ell_2} \lambda_{\ell_1} \lambda_{\ell_2} \sum_{\substack{k \\ (k,h)=s}} \beta_I(k)^2 r_R(n_1) r_R(n_2) \mathcal{W}_{\ell_1, \ell_2}(k), \quad (61)$$

$$n_i := \ell_i k + h,$$

and let $\mathcal{C}_{I,s}^{MM,\text{off}}$ be the same expression with $r_R(n_1) r_R(n_2)$ replaced by $\Lambda_R(n_1) \Lambda_R(n_2)$.

Theorem 7.3 (Nonprimitive finite-model reduction). *For each fixed divisor $s > 1$ of h and every $\varepsilon > 0$,*

$$\boxed{\mathcal{C}_{I,s}^{\text{off}} = \mathcal{C}_{I,s}^{MM,\text{off}} + O_{\varepsilon,h,W,\psi}(LX^\varepsilon).} \quad (62)$$

The coefficient in the finite-model correlation is endpoint-localized by (59)–(60).

Proof. Expand

$$\begin{aligned} r_R(n_1) r_R(n_2) &= \Lambda(n_1) \Lambda(n_2) - \Lambda(n_1) \Lambda_R(n_2) - \Lambda_R(n_1) \Lambda(n_2) \\ &\quad + \Lambda_R(n_1) \Lambda_R(n_2). \end{aligned}$$

Choose a prime $p \mid s$. Since $p \mid k$ and $p \mid h$, both n_i are divisible by p . If $\Lambda(n_i) \neq 0$, then $n_i = p^\nu$. On the fixed multiplicative support $n_i \asymp_W X$, there are $O_{h,W}(1)$ possible exponents ν . The equation

$$\ell_i k = p^\nu - h$$

has $O_\varepsilon(X^\varepsilon)$ factor pairs. Once one target von Mangoldt factor is fixed in this way, the other source variable has $O(L)$ choices. The soft bounds for β_I and Λ_R , and all logarithmic weights, are absorbed into another X^ε . Thus every term containing at least one target Λ is $O_{\varepsilon,h,W,\psi}(LX^\varepsilon)$. Only the last finite-model product remains. $\square$

Remark 7.4. Endpoint localization does not imply that the endpoint correlation is small. It changes its algebraic type and removes the interior interval; an estimate for the remaining finite-model lattice sum is still needed.

8 The exact finite-model CRT kernel

The Ramanujan identity

$$c_q(n) = \sum_{u \mid (q,n)} u \mu(q/u)$$

gives the finite divisor expansion

$$\Lambda_R(n) = \sum_{\substack{u \mid n \\ u \leq R}} \lambda'_R(u), \quad \lambda'_R(u) = u \sum_{m \leq R/u} \frac{\mu(um) \mu(m)}{\varphi(um)}. \quad (63)$$

We derive the exact compatibility conditions for the small lcm row $q \mid k$ in (30).

Lemma 8.1 (Three-modulus compatibility). *Let $q, u, v \leq R$, and let $\ell_1, \ell_2 > R$ be primes. The system*

$$q \mid k, \quad u \mid \ell_1 k + h, \quad v \mid \ell_2 k + h \quad (64)$$

is soluble if and only if

$$\boxed{(q, u) \mid h, \quad (q, v) \mid h, \quad (u, v) \mid h(\ell_1 - \ell_2).} \quad (65)$$

When soluble, k occupies one residue class modulo $[q, u, v]$.

Proof. The sources are invertible modulo u and v . Thus (64) is the system

$$k \equiv 0 \pmod{q}, \quad k \equiv -h\bar{\ell}_1 \pmod{u}, \quad k \equiv -h\bar{\ell}_2 \pmod{v}.$$

The generalized Chinese remainder theorem says that pairwise compatibility is necessary and sufficient. Compatibility of the first congruence with the second and third gives the first two divisibilities in (65). If $g = (u, v)$, compatibility of the last two is $h(\bar{\ell}_1 - \bar{\ell}_2) \equiv 0 \pmod{g}$. Multiplication by the unit $\ell_1 \ell_2$ turns this into $g \mid h(\ell_1 - \ell_2)$. The final statement is the uniqueness part of the generalized CRT. $\square$

Let $F \in C_c^1((0, \infty))$ and define the finite-model row

$$\mathcal{M}_R(q; \ell_1, \ell_2; F) = \sum_{\substack{k \geq 1 \\ q \mid k}} \Lambda_R(\ell_1 k + h) \Lambda_R(\ell_2 k + h) F(k/K). \quad (66)$$

Theorem 8.2 (Explicit two-target density). *One has the exact expansion*

$$\begin{aligned} \mathcal{M}_R(q; \ell_1, \ell_2; F) &= \sum_{u, v \leq R} \lambda'_R(u) \lambda'_R(v) \mathbf{1}_{(q, u) \mid h} \mathbf{1}_{(q, v) \mid h} \mathbf{1}_{(u, v) \mid h(\ell_1 - \ell_2)} \\ &\quad \times \sum_{k \equiv \kappa(q, u, v) \pmod{[q, u, v]}} F(k/K), \end{aligned} \quad (67)$$

where $\kappa(q, u, v)$ is the compatible class and an incompatible summand is zero. Its complete-period density, normalized relative to the q -row, is

$$\boxed{\rho_{h, R}^{(2)}(q; \ell_1, \ell_2) = q \sum_{u, v \leq R} \lambda'_R(u) \lambda'_R(v) \frac{\mathbf{1}_{(q, u) \mid h} \mathbf{1}_{(q, v) \mid h} \mathbf{1}_{(u, v) \mid h(\ell_1 - \ell_2)}}{[q, u, v]}.} \quad (68)$$

Consequently,

$$\mathcal{M}_R(q; \ell_1, \ell_2; F) = \frac{K}{q} \rho_{h, R}^{(2)}(q; \ell_1, \ell_2) I_0(F) + \mathcal{E}_R(q; \ell_1, \ell_2; F), \quad (69)$$

with the elementary bound

$$|\mathcal{E}_R(q; \ell_1, \ell_2; F)| \ll_F R^2 (\log(2R))^4. \quad (70)$$

Proof. Insert (63) twice in (66) and apply lemma 8.1. This proves the exact formula. On a compatible residue class modulo $M = [q, u, v]$, Euler summation gives

$$\sum_{k \equiv \kappa \pmod{M}} F(k/K) = \frac{K}{M} I_0(F) + O_F(1).$$

The main terms assemble into (68).

The formula (63) and the standard bound $n/\varphi(n) \ll \log \log(3n)$ give $|\lambda'_R(u)| \ll (\log(2R))^2$ uniformly. Summing the $O_F(1)$ errors over $u, v \leq R$ proves (70). $\square$

The third condition in (65) contains the transverse determinant $h(\ell_1 - \ell_2)$, whereas the first two retain the lcm row $q \mid k$. This is why replacing the geometry by the single product $hq(\ell_1 - \ell_2)$ loses information.

The exact kernel is useful, but the crude error in (70) is far too large after summing the two source variables. Controlling incomplete periods requires genuine cancellation in the u, v coefficients or a spectral/dispersion transformation. We claim no asymptotic evaluation of this kernel here.

9 The stopping dichotomy and a conditional closure gate

We assemble the two quadraticizations. For each positive divisor $s \mid h$, put

$$\mathcal{T}_s^{\mathbb{P}} = \sum_{\substack{k \\ (k,h)=s}} \beta_I(k) S_k. \quad (71)$$

Write $\mathcal{C}_{D,\text{prim}}^{\text{gen}}$ for the generic subsum $\mathcal{C}_D^{\text{gen}}$ of equations (41) and (49) with the fixed mask $\xi(d, j) = \mathbf{1}_{(dj,h)=1}$.

Theorem 9.1 (Primitive/nonprimitive stopping dichotomy). *Assume (5), $D_0 < V/2$, and $L \geq X^{\lambda_0}$ for a fixed $\lambda_0 > 0$. Suppose that, for fixed $a \geq 0$ and $\eta > 0$,*

$$|\mathcal{T}_{L,K}(I)| \geq \frac{\eta X}{(\log X)^a}. \quad (72)$$

Then, for every fixed $\kappa > 0$ and all sufficiently large X , at least one of the following alternatives holds.

- (A) **Primitive generic determinant witness.** *There is a dyadic $D \leq V$, with $Q = LD$, such that the primitive d -slice has a generic correlation satisfying*

$$\mathcal{C}_{D,\text{prim}}^{\text{gen}} \gg_{\eta,h,W} \frac{XQ}{(\log X)^{2a+2}}, \quad (73)$$

supported on

$$\ell_1 \neq \ell_2, \quad |\ell_1 d_1 - \ell_2 d_2| > QX^{-\kappa}, \quad (d_1, d_2) \leq X^\kappa, \quad (d_1 j, h) = (d_2 j, h) = 1.$$

- (B) **Nonprimitive endpoint finite-model witness.** *There is a divisor $s > 1$ of h for which*

$$\mathcal{C}_{I,s}^{MM,\text{off}} \gg_{\eta,h,W} \frac{XL}{(\log X)^{2a}}. \quad (74)$$

Its Möbius coefficients are supported only on the two endpoint bands in (60). After a dyadic split of $|\ell_1 - \ell_2|$, one signed separation cell contributes

$$\gg_{\eta,h,W} \frac{XL}{(\log X)^{2a+1}}. \quad (75)$$

Proof. By lemma 2.2, the source prime-power error is smaller than (72) by a fixed power. Since (k, h) can take only $O_h(1)$ values, one sector in (71) has magnitude $\gg_{\eta,h} X(\log X)^{-a}$.

If the large sector is $s = 1$, open $\beta_I(k)$ and use a smooth dyadic partition on $d \in I$. There are $O(\log X)$ slices, so one primitive slice (36), with $\xi(d, j) = \mathbf{1}_{(dj,h)=1}$, has magnitude $\gg X(\log X)^{-a-1}$. Apply theorem 6.3 with a replaced by $a + 1$ to obtain (73).

If the large sector has $s > 1$, repeat the same- k Cauchy argument of [theorem 5.1](#) with the restriction $(k, h) = s$. Its diagonal is still $O_\varepsilon(X^{1+\varepsilon})$, so

$$\mathcal{C}_{I,s}^{\text{off}} \gg_{\eta,h,W} XL(\log X)^{-2a}.$$

The error in (62) is negligible at this scale, which proves (74). Endpoint localization follows by squaring (59). There are $O(\log X)$ nonempty dyadic source-separation ranges; since their signed sum is positive and has the displayed size, at least one proves (75). $\square$

Remark 9.2 (Block scope). [Theorem 9.1](#) is attached to a symmetric tail block for which the TPC-17 prefix theorem applies. It does not assert that every possible failure of a fixed-shift prime-pair asymptotic must occur in this particular family of blocks. That global statement requires the full preceding normal-form interface.

We now isolate the exact analytic input that would remove alternative (A). This input is deliberately stated rather than assumed true.

Conditional input 9.3 (Generic determinant dispersion $\text{DD}_h(\theta)$). Fix $\theta > 1/2$. For every fixed $\varepsilon_0 > 0$, uniformly for every relevant slice (36) with $Q = LD \leq X^{\theta-\varepsilon_0}$, every fixed $\kappa > 0$, every admissible fixed-residue mask ξ described in [section 6](#), and every $A > 0$, one has

$$|\mathcal{C}_D^{\text{gen}}| \ll_{A,\varepsilon_0,\kappa,h,W,\psi} XQ(\log X)^{-A}. \quad (76)$$

Proposition 9.4 (Conditional slice and tail closure). *Assume the block geometry (5), so in particular $L \geq X^{\lambda_0}$ for some fixed $\lambda_0 > 0$, and assume $\text{DD}_h(\theta)$ covers every member of the fixed bounded-overlap dyadic partition of a given symmetric tail block. Then, for every $B > 0$,*

$$\mathcal{T}_{L,K}(I) \ll_B X(\log X)^{-B}. \quad (77)$$

Proof. Combine [theorem 6.2](#) and [Conditional input 9.3](#) with (39). For any requested B , take the logarithmic saving in (76) sufficiently large. Since $J \asymp X/Q$,

$$\begin{aligned} |\mathcal{T}_D| &\ll \sqrt{JX^{1+\varepsilon}} + \sqrt{JXQ}(\log X)^{-A/2} \\ &\ll XQ^{-1/2}X^{\varepsilon/2} + X(\log X)^{-A/2}. \end{aligned}$$

The first term has a fixed power saving because $Q \geq L \geq X^{\lambda_0}$. Sum the $O(\log X)$ divisor slices and restore the negligible source prime powers using [lemma 2.2](#). $\square$

9.1 How strong is the conditional input?

Write

$$L = X^{\lambda+o(1)}, \quad V = X^{v+o(1)}, \quad v = \frac{1}{4} - \frac{\delta}{2}.$$

The largest opened row has $Q = LV = X^{\lambda+v+o(1)}$. Define the boundary exponent

$$\theta_{\text{tail}} = \lambda + \frac{1}{4} - \frac{\delta}{2}. \quad (78)$$

Because [Conditional input 9.3](#) retains a fixed ε_0 margin, full-tail closure requires $\theta > \theta_{\text{tail}}$ by a fixed amount; equality is only the boundary value. For the two explicit TPC-17 profiles, let $\sigma > 0$ denote the fixed power-saving slack, subject to $\sigma < 1/100$ in the published row and $\sigma < 1/1000$ in the version-locked Li row. The boundary ledger is:

profile	λ and allowed δ	range of θ_{tail}
published Maynard layer	$10/21 - \sigma, 1/42 + 3\sigma < \delta < 1/4$	$101/168 - \sigma$ to $5/7 - (5/2)\sigma$
Li v6 extension	$8/17 - \sigma, 1/34 + 2\sigma < \delta < 1/4$	$81/136 - \sigma$ to $12/17 - 2\sigma$

The endpoints in the table are limiting open endpoints and are not attained by the displayed strict δ ranges. Numerically the lower endpoints are about 0.6012 and 0.5956 when σ is suppressed, while the upper endpoints are about 0.7143 and 0.7059. The Li row is version-locked to the unrefereed preprint used by TPC-17; the published theorem chain uses only the first row.

Maynard’s published well-factorable benchmark reaches level 3/5 for its specified coefficient class [10]. Lichtman’s preprint reaches $66/107 \approx 0.61682$ for triply well-factorable weights, with a conditional 5/8 statement under an additional spectral conjecture [7]. The numerical overlap with the lower end of (78) does not imply applicability: the sharp tail fails the balanced factor shape noted after [proposition 4.3](#), and (76) is a two-residual variable-determinant correlation rather than a one-prime progression estimate. Exponent size is not the missing hypothesis.

For clarity, the prime-pair singular series below is

$$\mathfrak{S}(h) = \prod_p \frac{1 - \nu_h(p)/p}{(1 - 1/p)^2}, \quad \nu_h(p) = \#\{a \bmod p : a(a+h) \equiv 0 \pmod{p}\}. \quad (79)$$

In particular it vanishes for odd h and is positive for nonzero even h .

Corollary 9.5 (Why uniform closure reaches the fixed-shift problem). *Assume the symmetric normal form of TPC-16 [14] and, as an additional global cover hypothesis, that its hard packet admits*

$$\mathcal{H}_{h,R;V}^{\text{HB}}(X) = \sum_{b \in \mathcal{B}_X} (\mathcal{C}_b + \mathcal{T}_b) + o(X), \quad |\mathcal{B}_X| = O_W(\log X), \quad (80)$$

where every block satisfies (5), the TPC-17 assembled-prefix estimate for $\mathcal{C}_b$, and the opened-d geometry for $\mathcal{T}_b$. Assume also $\text{DD}_h(\theta)$ with a fixed strict margin above every boundary in (78). Then the resulting chain gives the weighted fixed-shift Hardy–Littlewood asymptotic

$$\sum_n \Lambda(n) \Lambda(n+h) W(n/X) = \mathfrak{S}(h) X I_0(W) + o(X). \quad (81)$$

For $h = 2$ and nonnegative nonzero W , this would imply infinitely many twin primes after the standard removal of prime powers.

Proof. The assumed all-block prefix estimates remove the $\mathcal{C}_b$ terms. Proposition 9.4 removes each complementary $\mathcal{T}_b$. The global cover (80) and the TPC-16 normal form identify the sum of those blocks with the difference between the left side of (81) and its calibrated singular-series main term. There are only $O_W(\log X)$ blocks, so arbitrary logarithmic saving survives their sum. The $h = 2$ consequence follows from positivity and the fact that source or target prime powers contribute $o(X)$. $\square$

This corollary is a logical strength test, not a proof of its hypotheses. It explains why the primitive generic channel should not be dismissed as a routine Kloosterman estimate. By contrast, Matomäki et al. [8] establish the expected prime correlations for almost all shifts in sufficiently long ranges. That is a rigorous averaged- h benchmark, but it supplies no direct fixed- h , lcm-row-resolved estimate for (76).

10 Finite certificate and conclusion

The accompanying script `experiments/tpc18_certificate.py` performs exact finite checks of the algebraic layer. It verifies:

1. the lcm transform (11), its Möbius inversion, and the g, a, b parameterization;

2. the Gram identity (20) using rational arithmetic and the exact finite-box identity

$$\sum_{k \leq K} \beta_I(k)^2 = \sum_q \Gamma_I(q) \left\lfloor \frac{K}{q} \right\rfloor;$$

3. the terminal-octave lower bound and a numerical partial Euler product for C_{sf} ;
4. endpoint collapse for several nonprimitive contents;
5. the three CRT compatibility conditions by exhaustive residue enumeration;
6. a finite matrix TT^* identity and elementary tuple counts for the same-source, near-determinant, and large-gcd channels.

The generated JSON file records all parameters and results. These checks are deterministic regression tests for exact identities. They are not numerical evidence for a prime-pair asymptotic, for (76), or for the twin-prime conjecture.

10.1 What moved forward

TPC-17 ended with a sharp Möbius tail but no quadratic form adapted to that coefficient. The present paper supplies two such forms and proves that they reveal different, compatible information:

$$\begin{aligned} \text{same-}k \text{ square} &\implies q = [d_1, d_2] \leq V^2 \leq R, \quad \text{explicit lcm/CRT cells,} \\ d\text{-first square} &\implies m_i = \ell_i d_i > R, \quad h(m_1 - m_2), \text{ generic determinant channel.} \end{aligned} \tag{82}$$

The first route also proves a coefficient no-go result: the terminal octave carries nonvanishing Gram mass. The second route removes three large but elementary degeneracies. The content split then shows that nonprimitive sectors are not simply smaller copies of the primitive problem: their interior tail cancels exactly and their target-prime terms collapse to sparse prime powers.

The remaining primitive large-determinant/low-gcd correlation is a sharper and more testable gate than the undifferentiated tail. It is also honestly hard: a uniform bound at all necessary row exponents, combined with the preceding papers, reaches the fixed-shift Hardy–Littlewood asymptotic. Future progress can therefore proceed in three falsifiable directions:

- exploit additional factor shapes inside the generic channel without claiming coverage of the sharp tail as a whole;
- transform the variable determinant through dispersion, reciprocity, Poisson summation, and a Kuznetsov-type spectral step, with every incomplete-period error retained;
- first prove an averaged-shift or restricted-determinant theorem, then state precisely what extra uniformity is missing at fixed h .

Any one of these may yield a partial positive theorem or a rigorous no-go result. What has been gained here is a stable junction at which such alternatives no longer conceal one another. No parity barrier has been broken, but a broad unresolved packet has been reduced to two explicit and structurally different obstructions.

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